

A Pre-feasibility study for the establishment of an olive oil filling plant in Jerash



2017

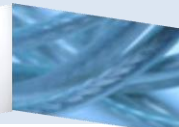


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1. Executive summary

تتلخص فكرة المشروع بتصنيع عبوات زجاجية شفافة وبتصاميم مميزة وبمختلف الأحجام إلى جانب وجود معصرة للزيتون العضوي وخط إنتاجي آخر لتعبئة الزيت المستخرج في عبوات زجاجية ليتم تغليفها ووضعها في صناديق كرتونية مناسبة.

The project idea summarized in manufacturing glass bottles with distinctive designs of various sizes, as well as an organic olive in addition to production line for filling the oil extracted in glass bottles to be packaged and placed in appropriate cartons.

تم تبني الفكرة لغنى المحافظة بالزيتون وزيت الزيتون مع تعبئته حالياً بالطرق التقليدية بعبوات حديدية مما يفقده إمكانية التصدير للأسواق الخارجية خاصة مع وجود طلب كبير على زيت الزيتون المصنع من الزيتون العضوي خارجياً.

يجدر بالذكر بأنه لضمان نجاح المشروع، فإنه يستلزم معرفة متطلبات السوق الخارجي من ناحية نسبة الحموضة المطلوبة ونوع الزيتون المطلوب استخدامه، ليتم تحديد مصادر الزيتون الذي سيتم استخدامه في عمليات إنتاج الزيت.

The idea was applied since the governorate is rich with olives and olive oil, and it is currently filled with traditional methods in iron containers, which makes it difficult to export to foreign markets, particu with a large demand for olive oil extracted from organic olive.

It is worth mentioning that to ensure the success of the project, it requires good knowledge of the requirements of the external market in terms of the required acidity and the type of olive to be used, to identify the sources of olives that will be used in the production of oil.

بناءً على الحصة السوقية المقترحة والأسعار التي تم اعتمادها للمشروع المقترح، بلغت الإيرادات المتوقعة 520,482 دينار أردني خلال السنة الأولى من عمر المشروع وترتفع تدريجياً لتصل إلى 679,111 دينار أردني خلال السنة العاشرة وتبلغ فترة استرداد مبلغ رأس المال 4 سنوات وبمعدل عائد داخلي يساوي 28.85%. بالإضافة إلى توفير 17 فرصة عمل لأبناء المنطقة.

Based on the proposed market share and the prices considered by the proposed project, the expected revenues amounted to 520,482 JD during the first year of the project life and gradually increase to reach 679,111 JD during the tenth year. The pay back period is 4 years and at an internal rate of return of 28.85%. In addition to providing 17 jobs for the residents of the governorate.

Table 1 economical indicators

Indicator	
Total Investment Cost (Jordan Dinars)	641,864
Internal Rate of Return (IRR)	%28.85

Pay-Back Period (Years)	4
Net Present Value (Jordan Dinars)	761,981
Job Opportunities provided by the Project	17
Total Revenues (For 10 Years)	5,966,742
Total Operating Expenses (For 10 years)	3,024,728
Total Net Profit (For 10 years)	1,008,938

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

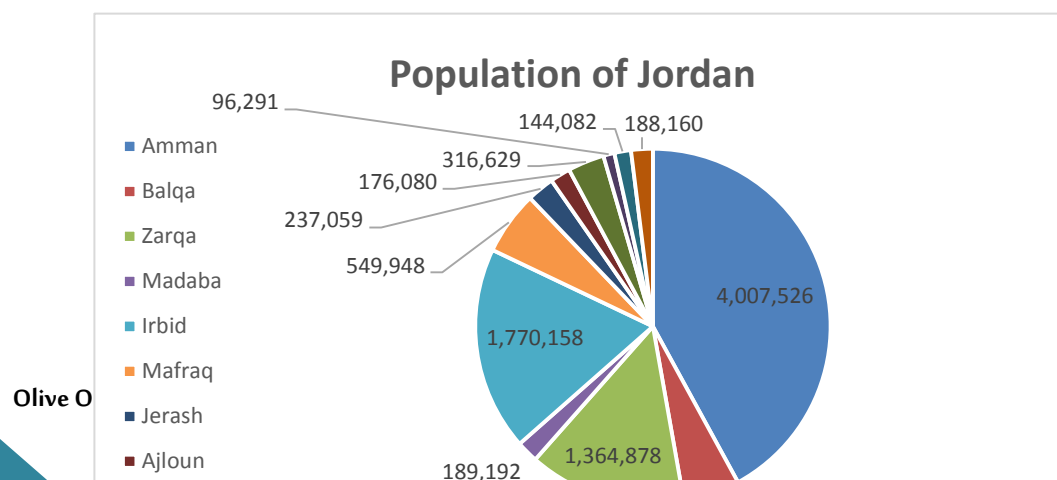
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

Department of Statistics (DOS), 2015.

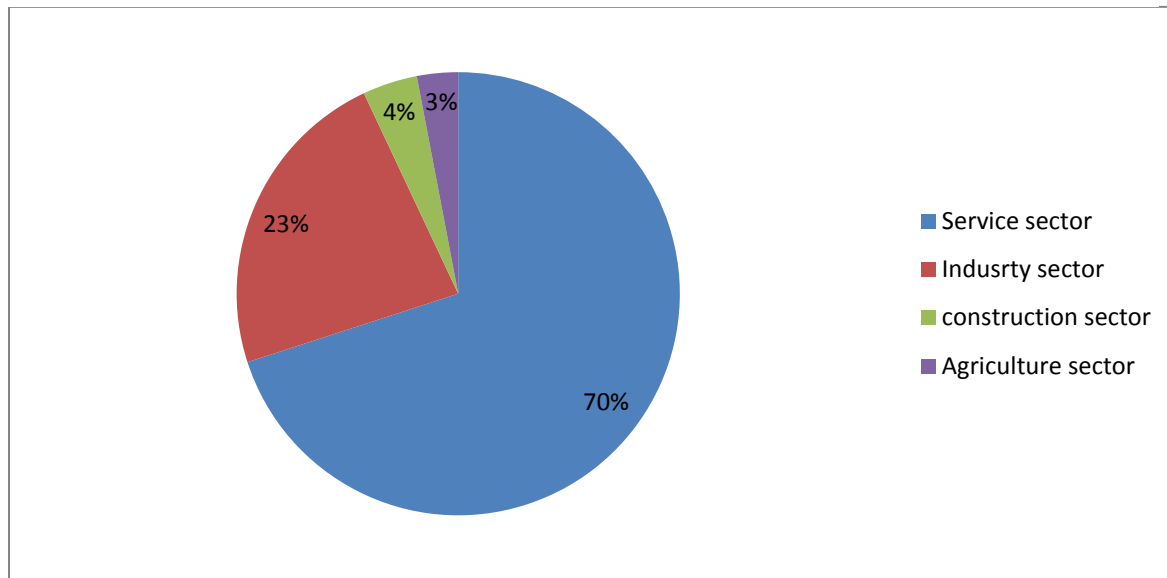
Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is

2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy

facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

2. Market analysis

Project description

تتلخص فكرة المشروع بإنشاء مصنع لتعبئة زيت الزيتون بعبوات زجاجية شفافة وبتصاميم مميزة إلى جانب وجود معصرة للزيتون العضوي ليتم تغليفها ووضعها في صناديق كرتونية متميزة.

The idea of the project is to establish a factory to fill olive oil in a transparent glass bottles with distinctive designs to be packaged and placed in a distinct cartons. In addition to the existence of an organic olive plant.

تم تبني الفكرة لغنى المحافظة بالزيتون وزيت الزيتون مع تعبئته حالياً بالطرق التقليدية بعبوات حديدية مما يفقده إمكانية التصدير للأسواق الخارجية خاصة مع وجود طلب كبير على زيت الزيتون المصنع من الزيتون العضوي خارجياً.

يجدر بالذكر بأنه لضمان نجاح المشروع، فإنه يستلزم معرفة متطلبات السوق الخارجي من ناحية نسبة الحموضة المطلوبة ونوع الزيتون المطلوب استخدامه، ليتم تحديد مصادر الزيتون الذي سيتم استخدامه في عمليات إنتاج الزيت.

تقسم أنواع الزيت في الأردن حسب نسبة الحموضة إلى الآتي:

The idea was adapted since the governorate is with olives and olive oil, and it is currently filled with traditional methods in iron containers, which makes it difficult to export to foreign markets, taking into account the large demand for olive oil made from organic olive.

It is worth mentioning that to ensure the success of the project, it requires good knowledge of the requirements of the external market in terms of the required acidity and the type of olive to be used, to identify the sources of olives that will be used in the production of oil.

The types of oil in Jordan are divided according to the percentage of acidity as follows:

- الزيت البكر الممتاز: لا تزيد نسبة حمض اولييك على 0.8 غ / 100 غ
- زيت الزيتون البكر: لا تزيد نسبة حمض اولييك على 2 غ / 100 غ
- زيت الزيتون البكر العادي: حمض اولييك 3.3 غ / 100 غ

- Extra oil: Oleic acid should not exceed 0.8 g / 100 g
- Extra olive oil: Oleic acid should not exceed 2 g / 100 g
- Average olive oil: Oleic acid 3.3 g / 100 g

مما يعني ان نسبة الحموضة يجب ان لا تزيد على 3.3 في الاردن. أما بالنسبة لمواصفات زيت الزيتون بالنسبة لوزارة الزراعة في الولايات المتحدة (USDA) فيجب ان لا تزيد نسبة حمض اولييك عن 2 غ / 100 غ.

Which means that the acidity should not exceed 3.3 in Jordan. As for the specifications of olive oil for the United States Department of Agriculture (USDA), the percentage of the oleic acid should not exceed 2 g / 100 g.

سيتم اختيار مزارع زيتون خاصة في محافظة جرش تضمن توفير زيتون عضوي لعصره داخل المعصرة وإنتاج زيت زيتون عضوي بالمواصفات المطلوبة. سيتكون المصنع من جزئين، جزء خاص بعصر الزيتون العضوي واستخراج الزيت بنسبة حمضية لا تتعدى عن 0.8 غ / 100 غ، حيث يتم فحص عينات منه بعد العصر لضمان المحافظة على نسبة الحموضة وجودة المنتج المقدم ومن ثم تخزينه حسب المواصفات والمقاييس الخاصة بتخزين زيت الزيتون ومن ثم تنقيته بالطرق الفيزيائية لحين تعبئته، وتفريغه من خلال خط الإنتاج في عبوات خاصة، أو تعبئته فوراً في حال توافر العبوات الزجاجية، ومن ثم تسويقه إلى الفئات المستهدفة.

An olive plantation will be selected in Jerash governorate, which will provide organic olive to be extracted inside the olive plant and produce organic olive oil according to the required specifications. The plant will consist of two parts, A special part of the organic olive oil and oil extraction with acidity not exceeding 0.8 g / 100g, where samples will be examined after being extracted to ensure the acidity percentage and quality of the product then stored according to the specifications and standards for olive oil storing and then Purification by physical methods until filling, unloading through the production line in special packaging, or filling it immediately in case of glass containers are available at the plant to be distributed to the targeted segments.

Project Objective

تتلخص أهداف المشروع بالنقاط التالية:

- تحقيق عائد ربحي للجهة القائمة على المشروع.
- توفير زيت زيتون بجودة عالية وعبوات أقل حجما من الأحجام المتعارف عليها.
- استغلال الفائض من زيت الزيتون.
- النهوض بالقطاع الزراعي في محافظة جرش.
- زيادة الاستثمار في جرش.
- توفير فرص عمل لآبناء محافظة جرش لتقليل البطالة في المنطقة.

The objectives of the project are summarized as follows:

- Achieve a profitable return to the owners.
- Provide a high quality of olive oil with smaller packaging sizes than conventional sizes.
- Exploit the excess of olive oil.
- Improving the agricultural sector in Jerash.
- Increase investments in Jerash.
- Provide job opportunities for the residents of Jerash governorate.

Proposed product

إن المنتج النهائي المراد طرحه في الأسواق وتقديمه للمستهلكين هو زيت الزيتون البكر الممتاز المعبأ في عبوات زجاجية، وبحجم 1 لتر من زيت الزيتون.

The final product presented to consumers is olive oil packed in glass containers, with a volume of 1 liter of olive oil.

Targeted segments

يستهدف المشروع المقترح الفئات التالية:

- السوق المحلي: المجتمع الأردني.
- السوق الخارجي: التصدير للدول الخارجية.

The proposed project targets the following segments:

- Local market: Jordanian society.
- External market: export to foreign countries.

Market size

تعد الاردن من أكثر الدول انتاجية لزيت الزيتون حيث انتجت عام 2015 نحو 29.611 ألف طن، وبمتوسط انتاجي في محافظة جرش من زيت الزيتون ما يقارب 3300 طن. كما أن معدل استهلاك الفرد الواحد من زيت الزيتون يساوي 3.2 كغ في السنة وعيله فإن كمية الاستهلاك لزيت الزيتون لعام 2017 سيكون حوالي 31.407 ألف طن.

Jordan is one of the most productive countries of olive oil, which produced in the year 2015 about 29.611 thousand tons, and the average production in Jerash of olive oil is about 3300 tons. The average per capita consumption of olive oil is 3.2 kg per year, and the consumption of olive oil for 2017 will be about 31,407 thousand tons.

1.1 The analysis of competitors

1. Al Tuor for olive oil products

يقوم مصنع الطور لمنتجات زيت الزيتون في محافظة معان ببيع المنتجات من زيت الزيتون بالأحجام والأسعار التالية:
Al-Tuor factory for olive oil products in Ma'an Governorate is selling products of olive oil in the following sizes and prices:

Table 3 Selling prices of competitors (Al Tuor factory)

Pack	Selling Price (JD)
500 ml bottle	3
750 ml bottle	3.5

3 liter bottle

14

2. Al-Alia Company for Specialized Industries

تأسست شركة العليا للصناعات المتخصصة في عام 2001 وهي شركة متخصصة في إنتاج زيت البكر الممتاز وتنتج أيضا أنواع أخرى من الزيت مثل زيت الصويا وزيت عباد الشمس ويقوم المصنع بتوفير المنتجات التالية:

Alia Industries was established in 2001 and specialized in producing premium oil, other types of oil such as soybean oil and sunflower oil. The factory is providing the following products:

Table 4 Prices of competitors (Al-Alia Company for Specialized Industries)

Pack	Selling Prices (JD)
1 liter plastic pack	4.95
16 kg pack	60
10 kg pack	40
4 liter pack	19
2 liter pack	10
2.7 pack	12.75
750 ml bottle	4
700 ml bottle	3.7
500 ml bottle	2.85
250 ml bottle	2.45
700 ml plastic pack	3.7
500 ml plastic pack	2.55

3. Al Mutakamilah Company for olive products

تأسست الشركة المتكاملة لمنتجات الزيتون في عام 2001 ويقع مصنعها في محافظة عجلون ويقوم ببيع المنتجات التالية:

The company was established in 2001 and its factory is located in Ajloun Governorate. The following table should its products:

Table 5 Prices of competitors (Al Mutakamilah Company for olive products)

Pack	Selling Price (JD)
16 liter pack	70
4 liter pack	19.35

2.7 liter pack	13.30
2 liter pack	10.35
750 ml bottle	4.5
500 ml bottle	3
250 ml bottle	1.7

Market share

The following table shows the productive capacity of the plant:

Table 6 Expected production capacity

Analysis of production capacity	
Average production of Jarash Governorate of olive oil in tons	3300
Percentage of the targeted quantity of production in Jerash Governorate	1.50%
Expected production quantity in kg	49500
Conversion coefficient from kilogram to liter	0.913

* معامل التحويل من كيلو غرام إلى لتر أقل من معامل تحويل الماء (كيلو غرام = لتر) لأن كثافة الزيت أقل من كثافة الماء.

* The conversion coefficient from kg to L is lower than the water conversion factor (kg = L) because the oil density is less than the water density

Price analysis and pricing policy

Based on similar product prices, the suggested prices are:

- 1 liter local price: 8 JD
- 1 liter export price: 12 JD

Local sales are expected to be 60% and 40% for export.

Projected revenues

Table 7 projected revenues

Income sources	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
The price of selling a 1 liter locally	8.00	8.24	8.49	8.74	9.00	9.27	9.55	9.84	10.13	10.44
Selling price of 1 liter externally	12.00	12.36	12.73	13.11	13.51	13.91	14.33	14.76	15.20	15.66
Local sales ratio	60 %	60%	60%	60%	60%	60%	60%	60%	60%	60%
Export Ratio	40%	40%	40%	40%	40%	40%	40%	40%	40%	40%
Number of 1 liter produced bottles	54217	54217	54217	54217	54217	54217	54217	54217	54217	54217
Revenues	520482	536096	552179	568745	585807	603381	621483	640127	659331	679111

3. Technical analysis

The required human resources

Table 8 the required man power for the project

Job title	No.
General Manager	1
Marketing manager	1
Accountant	1
Sales Representative	2
Maintenance Officer	1
Guard	1
Cleaner	1
Total indirect staff	8
Director of Production Department	1
Production worker	5
Quality control	1
Driver	2
Total direct staff	9
Total Staff	17

وفي ما يلي توضيح لوظائف القوى العاملة:

- **مدير عام:** يعمل منفردًا وضمن الصلاحيات المخولة له ويقوم باقتراح سياسات وخطط العمل، وتخطيط وإدارة سير العمل بالمشروع، ومتابعة تنفيذ الخطط والبرامج بما يتفق والسياسات المقررة وأهداف وغايات المشروع بالشكل الأمثل، والإشراف على إدارة الشؤون الإدارية والمالية في المشروع بما ينسجم والتعليمات المقررة بهذا الخصوص، وإدارة المرؤوسين وتنمية مهاراتهم، وتطوير إجراءات وتأمين مستلزمات السلامة والصحة المهنية والعامة.
- **General Manager:** works within the authority vested to him and proposed policies, work plans, planning and managing the work, follow-up implementation of plans and programs in accordance with the policies and goals and objectives in an optimal manner, and oversee the administrative and financial management. In addition to the management of subordinates and the development of their skills, the development of procedures and ensure the requirements of occupational and public safety and health.

- **مدير إنتاج:** متابعة ومراقبة سير الإنتاج حسب الخطط الموضوعة مسبقا.

- **Production Manager:** Follow-up and monitoring of the production process according to pre-set plans.
- محاسب: متابعة الأمور المالية والتدقيق المالي، وتقديم التقرير السنوي للوضع المالي.
- **Accountant:** follow-up financial matters and financial audit, and submit annual reports of the financial situation.
- موظف تسويق: يقوم بدراسة احتياجات السوق، ومتابعة نشاطات الدعاية والإعلام وتقييم فاعليتها، ومراقبة مستوى المبيعات وتنظيمها.
- **Marketing Officer:** examines the needs of the market, and follow up the activities of publicity and evaluation of effectiveness, and monitors the level of sales.
- موظف الصيانة: يقوم بإجراءات ونشاطات الصيانة الكهربائية والميكانيكية، ومتابعة تنفيذ الخطط والبرامج المقررة بما يكفل استمرارية العملية الإنتاجية وجودتها بأقصى طاقة وأقل كلفة ممكنة، ومتابعة توفير قطع الغيار اللازمة، وإعداد التقارير الفنية.
- **Maintenance Officer:** Performs the electrical and mechanical maintenance procedures and activities, and follow up the implementation of the plans scheduled to ensure continuity of the production process with maximum capacity and lowest possible cost. In addition of preparing technical reports.
- عامل فني: يقوم بتشغيل خط الإنتاج وتحميل وتنزيل البضائع والمواد الأولية.
- **Technical worker:** responsible of the production line, loading and unloading of products and raw materials.
- مندوب مبيعات: يعمل منفرداً ضمن الصلاحية المخولة له ويقوم بإعداد خطط وبرامج العمل وتنظيم وتنسيق العمليات الإجرائية، والعمل على توثيق العلاقة بين المؤسسة والمؤسسات المختلفة أو الفئات التي تتعامل معها عبر وسائل الإعلام والتوعية المختلفة، وإعداد التقارير الفنية، وإدارة المرؤوسين وتنمية مهاراتهم.
- **Salesman:** works within the authority vested to him, prepares work plans and programs, organizes and coordinates the procedural processes, and works to document the relationship between the company and other different companies or groups that deal with them. In addition of preparing technical reports, managing subordinates and developing their skills.
- مراقب جودة: التحقق من فاعلية الاداء الاداري واجراءات العمل وفق التشريعات المعمول بها.
- **Quality Control:** verifying the effectiveness of administrative performance and work procedures in accordance with the legislation.

Site analysis

يقع المكان التنفذي للمشروع في محافظة جرش، بالقرب من مزرعة الزيتون العضوي الموردة للزيتون لتقليل نفقات النقل بين المورد والمصنع، لذلك يعتمد موقع المصنع على موقع مزرعة الزيتون العضوي المراد التعاقد معها.

The project is located in Jerash Governorate, close to the organic olive farm that supplies the olives in order to reduce the transportation expenses between the supplier and the factory. Therefore, the plant site depends on the location of the organic olive farm to be contracted with.

Description of the production process

تتلخص خطوات التصنيع في النقاط التالية:

- تنقية ثمار الزيتون من كل ما يعلق بها من شوائب.
- غسل الزيتون بالماء الساخن.
- مرحلة العصر من خلال وضعه بالة العصر لاستخراج الزيت منه.
- اخذ عينات من الزيت وفحص درجة حموضته.
- حفظ الزيت في خزانات خاصة لحين تعبئته.
- تعبئة الزيت من خلال الالة المخصصة في عبوات زجاجية لتقديمها للمستهلك.

Manufacturing steps are summarized as follows:

- Purify olives from all impurities attached.
- Wash olives with hot water.
- The pressing stage through pressing machine used to extract oil from olives.
- Taking oil samples and checking the pH of oil.
- Keep oil in special tanks until filling.
- Filling oil through a machine in glass bottels to be delivered to the consumer.

The analysis of technical requirements

1.1.1 Land and construction

The project needs a total area of 1000 m². The following table shows the required land for the proposed project

Table9 the required land for the project

Land	Area sq.m
------	-----------

The required land **1,000**

Source: Directorate of registration: Ma'an

The following table shows the required construction work for the project:

Table 10 the required construction work

Construction work	Area sq.m
Henger	800
Paving	800
Land settlement works	800
Building a fence	140
Other works	-

Machinery and Equipment

The following table shows the required machinery and equipment needed for the project:

Table 11 Machinery and equipments

Machinery and equipment	Quantity
Production line	1
Olive oil storage tanks	20
Olive oil filter	1
Tank	1
pumps	10
Laboratory examination device	1
Stainless steel pipes	
water tanks	3
Presser	1

Vehicles and modes of transportation

The following table shows a list of vehicles and modes of transportation required for the project

Table 12 Vehicles and modes of transportation

Vehicles and modes of transportation	Quantity
Pickup truck	2

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.88% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 8 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 13 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 14 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 15 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 16 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	25.00%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	1.50%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.50%

Table 17 Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	20%
Income Tax Deduction(2)	60%
Income Tax after Deduction	8.00%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	
Annual Monthly Salaries have been calculated after multiplying monthly	16

Risk Premium	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	9.48%
Market Risk Premium (3)	10.12%
Income Tax Rate	8.00%
Beta (4)	1
Growth rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 18 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%8
Bets	1.00
Equity	256,746
Loans Value	385,118
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%8.72
Loans Value	385,118
Loans Percentage	%60
Equity	
Cost of Equity	%16.6
Equity Value	256,746
Equety Percentage	%40
Gross	
Project Value	641,864
Proportion of the project	%100
Weighted Average Cost of Capital	%11.88

Capital Expenditures

The estimated cost of the project is 618,712 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 19 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square)	Cost	Area (M2)	Cost
Required Land	7		4,000	28,000
Total				28,000

Table 20 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost Jordan /
Henger	100	800	80,000
Paving	25	800	20,000
Land settlement works	30	800	24,000
Building a fence	70	140	9,800
Other works			6,690
Total			140,490

Table 21 Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost / JD	Total Cost
Production line	1	30,000	30,000
Olive oil storage tanks	20	10,000	200,000
Olive oil filter	1	8,000	8,000
Tank	1	500	500
pumps	10	350	3,500
Laboratory examination device	1	500	500
Stainless steel pipes			2,500
water tanks	3	350	1,050
Presser	1	15,000	15,000
Total			261,050

Table 22 Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Furniture	-	-	7,000
Total			7,000

Table 23 Transporation and Vehicles Capex

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Pick up	2	23,400	46,800
Total			46,800

Table 24 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	3,000
Transportation	750
Marketing	2,500
Training	2,500
Total	8,750

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 25 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	64,998
General and administrative expenses	3	%25	2,726
Marketing expenses	3	%25	651
Indirect Salaries	3	%25	23,000
Total			91,375

Table 26 Expenses Summary

Capital expenditure	Cost (JD)	Contingency	Cost (JD)	%
Land	28,000	10%	30,800	%4.8
Construction works	140,490	10%	154,539	%24.1
Machinery &Equipment	261,050	10%	287,155	%44.7
Furniture	7,000	10%	7,700	%1.2
Vehicles	46,800	10%	51,480	%8.0
Pre-Operating Expenses	8,750	10%	9,625	%1.5
Working capital	91,423	10%	100,565	%15.7
Total (JD)	583,513		641,864	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (641,864) JD, where 60% of the project will be financed through loans (385,118) JD, and the remaining (40%) through (Equity) with (256,746) JD. The following table summarizes the general structure of the required financing:

Table 27 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	256,746	%40
Loans	385,118	%60
Total	641,864	%100
Use of Fund	Value	Ratio
Capital Expenditures	531,674	%82.83
Pre-Operating Expenses	9,625	%1.50
Working Capital	100,565	%15.67
Total	641,864	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 28 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials	130,120	134,024	138,045	142,186	146,452	150,845	155,371	160,032	164,833	169,778
Utilities	26,024	26,805	27,609	28,437	29,290	30,169	31,074	32,006	32,967	33,956
Maintenance	7,807	8,041	8,283	8,531	8,787	9,051	9,322	9,602	9,890	10,187
Disposables	5,205	5,361	5,522	5,687	5,858	6,034	6,215	6,401	6,593	6,791
Direct Salaries	67,200	69,955	72,823	75,809	78,917	82,153	85,521	89,028	92,678	96,477
Others	23,636	24,419	25,228	26,065	26,930	27,825	28,750	29,707	30,696	31,719
Total	259,992	268,605	277,510	286,716	296,235	306,077	316,253	326,776	337,656	348,907

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 29 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Stationary and Printing	600	618	637	656	675	696	716	738	760	783

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Transportaton	2,500	2,575	2,652	2,732	2,814	2,898	2,985	3,075	3,167	3,262
Training	600	618	637	656	675	696	716	738	760	783
Insurance	3,988	4,107	4,230	4,357	4,488	4,623	4,761	4,904	5,051	5,203
Miscellaneous	1,009	1,039	1,070	1,102	1,135	1,169	1,205	1,241	1,278	1,316
Total	11,096	11,429	11,772	12,125	12,489	12,864	13,250	13,647	14,056	14,478

Marketing Expenses

Table 30 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	2,602	2,680	2,761	2,844	2,929	3,017	3,107	3,201	3,297	3,396

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 31 Expected monthly salaries (2018-2027)

المسمى الوظيفي	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
General Manager	1,750	1,822	1,896	1,974	2,055	2,139	2,227	2,318	2,413	2,512
Marketing manager	1,200	1,249	1,300	1,354	1,409	1,467	1,527	1,590	1,655	1,723
Accountant	850	885	921	959	998	1,039	1,082	1,126	1,172	1,220
Salesman	450	468	488	508	528	550	573	596	621	646
Maintenance Officer	450	468	488	508	528	550	573	596	621	646
Guard	300	312	325	338	352	367	382	397	414	431
Cleaner	300	312	325	338	352	367	382	397	414	431
Direct staff										
Director of Production Department	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Production worker	350	364	379	395	411	428	445	464	483	502
Quality control	750	781	813	846	881	917	954	994	1,034	1,077
Driver	350	364	379	395	411	428	445	464	483	502

Table 31 Annual Expected Salaries

المسمى الوظيفي	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
General Manager	28,000	29,148	30,343	31,587	32,882	34,230	35,634	37,095	38,616	40,199

المسمى الوظيفي	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing manager	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Accountant	13,600	14,158	14,738	15,342	15,971	16,626	17,308	18,017	18,756	19,525
Salesman	14,400	14,990	15,605	16,245	16,911	17,604	18,326	19,077	19,860	20,674
Maintenance Officer	7,200	7,495	7,803	8,122	8,455	8,802	9,163	9,539	9,930	10,337
Guard	4,800	4,997	5,202	5,415	5,637	5,868	6,109	6,359	6,620	6,891
Cleaner	4,800	4,997	5,202	5,415	5,637	5,868	6,109	6,359	6,620	6,891
Total Indirect	92,000	95,772	99,699	103,786	108,042	112,471	117,083	121,883	126,880	132,082
Direct staff										
Director of Production Department	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Production worker	28,000	29,148	30,343	31,587	32,882	34,230	35,634	37,095	38,616	40,199
Quality control	12,000	12,492	13,004	13,537	14,092	14,670	15,272	15,898	16,550	17,228
Driver	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Total Direct	67,200	69,955	72,823	75,809	78,917	82,153	85,521	89,028	92,678	96,477
Grand Total	159,200	165,727	172,522	179,595	186,959	194,624	202,604	210,910	219,558	228,560

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 32 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	30,800	%0.0	%0.0
Construction Works	154,539	%5.0	%0.0
Machinaries	287,155	%10.0	%2.0
Furniture	7,700	%10.0	%5.0
Vehicles	51,480	%10.0	%0.0
	531,674		

Table 33 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	531,674	537,802	544,064	550,464	557,004	563,689	570,521	577,505	584,643	591,941
Total Depreciation Expenses	42,360	42,973	43,599	44,239	44,893	45,562	46,245	46,944	47,657	48,387
Total Accumulated Depreciation	42,360	85,334	128,933	173,173	218,066	263,628	309,873	356,817	404,474	452,861
Total Additions	0	6,128	6,262	6,400	6,540	6,684	6,832	6,984	7,139	7,298
Total Net Book Values	489,314	452,468	415,131	377,291	338,938	300,061	260,648	220,688	180,169	139,080

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 34 Loan Details

Loan Value	385,118
Annual Interest Rate	9.48%
Loans Period	5
Loan Starts at year:	2018
Annual Payment	100,247
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				385,118
2019	100,247	36,509	63,738	321,380
2020	100,247	30,467	69,780	251,600
2021	100,247	23,852	76,396	175,205
2022	100,247	16,609	83,638	91,567
2023	100,247	8,681	91,567	0

Income Statement

Table 35 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	520,482	536,096	552,179	568,745	585,807	603,381	621,483	640,127	659,331	679,111
Gross Operating Revenues	520,482	536,096	552,179	568,745	585,807	603,381	621,483	640,127	659,331	679,111
Operating Expenses	(259,992)	(268,605)	(277,510)	(286,716)	(296,235)	(306,077)	(316,253)	(326,776)	(337,656)	(348,907)
Gross Operating Profit	260,490	267,491	274,669	282,028	289,572	297,304	305,229	313,351	321,675	330,204
Gross Profit Percentage	%50	%50	%50	%50	%49	%49	%49	%49	%49	%49
Salaries and Benefits (Indirect Staff)	(92,000)	(95,772)	(99,699)	(103,786)	(108,042)	(112,471)	(117,083)	(121,883)	(126,880)	(132,082)
General and Administraive Expenses	(11,096)	(11,429)	(11,772)	(12,125)	(12,489)	(12,864)	(13,250)	(13,647)	(14,056)	(14,478)
Markeing Expenses	(2,602)	(2,680)	(2,761)	(2,844)	(2,929)	(3,017)	(3,107)	(3,201)	(3,297)	(3,396)
Pre Operating Expenses	(9,625)									
Gross Indirect Expenses	(115,324)	(109,882)	(114,232)	(118,755)	(123,460)	(128,352)	(133,440)	(138,731)	(144,233)	(149,956)
Income before Interest, Depreciation and Tax	145,166	157,610	160,438	163,273	166,112	168,952	171,790	174,621	177,441	180,248
Fixed Assets Depreciations	(42,360)	(42,973)	(43,599)	(44,239)	(44,893)	(45,562)	(46,245)	(46,944)	(47,657)	(48,387)
Income before Tax and Interests	102,805	114,636	116,838	119,034	121,219	123,391	125,545	127,677	129,784	131,860
Bank Interests	(36,509)	(30,467)	(23,852)	(16,609)	(8,681)	(0)	(0)	(0)	(0)	(0)
Income Before Tax	66,296	84,169	92,987	102,424	112,538	123,391	125,545	127,677	129,784	131,860

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Tax	(5,304)	(6,734)	(7,439)	(8,194)	(9,003)	(9,871)	(10,044)	(10,214)	(10,383)	(10,549)
Net Profit	60,993	77,436	85,548	94,230	103,535	113,519	115,501	117,463	119,401	121,312
<i>Net Profit Percentage</i>	<i>%12</i>	<i>%14</i>	<i>%15</i>	<i>%17</i>	<i>%18</i>	<i>%19</i>	<i>%19</i>	<i>%18</i>	<i>%18</i>	<i>%18</i>
Compulsory Reserves	(6,099)	(7,744)	(8,555)	(9,423)	(10,354)	(11,352)	(11,550)	(11,746)	(11,940)	(12,131)
Retained Earnings	54,893	124,586	201,579	286,386	379,568	481,735	585,686	691,403	798,864	908,044

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 36 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	60,993	77,436	85,548	94,230	103,535	113,519	115,501	117,463	119,401	121,312
Non-monetary adjustments										
Bank benefits	36,509	30,467	23,852	16,609	8,681	0	0	0	0	0
Consumption	42,360	42,973	43,599	44,239	44,893	45,562	46,245	46,944	47,657	48,387
Operating cash flows before changes in working capital	139,862	150,876	152,999	155,079	157,109	159,081	161,746	164,407	167,059	169,699
Shortage (increase) in inventory	(10,843)	(325)	(335)	(345)	(355)	(366)	(377)	(388)	(400)	(412)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
(Decrease) increase in accounts payable	21,645	785	814	844	876	908	942	977	1,013	1,051
Change in working capital	10,802	460	479	499	520	542	565	588	613	639
Net cash flows (used) from operating activities	150,664	151,336	153,478	155,578	157,629	159,623	162,311	164,995	167,672	170,338
Cash flows from investing activities										
(Purchase) of fixed assets	(531,674)	(6,128)	(6,262)	(6,400)	(6,540)	(6,684)	(6,832)	(6,984)	(7,139)	(7,298)
Net flows (used) from investment activities	(531,674)	(6,128)	(6,262)	(6,400)	(6,540)	(6,684)	(6,832)	(6,984)	(7,139)	(7,298)
Cash flows from financing activities										
Equity capital	256,746									
Loan Amortization	(63,738)	(69,780)	(76,396)	(83,638)	(91,567)	0	0	0	0	0
Bank Interest Rate	(36,509)	(30,467)	(23,852)	(16,609)	(8,681)	(0)	(0)	(0)	(0)	(0)
Loans	385,118	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	541,617	(100,247)	(100,247)	(100,247)	(100,247)	0	0	0	0	0
Net increase (decrease) in cash	160,607	44,960	46,968	48,931	50,842	152,939	155,479	158,011	160,533	163,040
Cashflows at the Beginning of Period	0	160,607	205,567	252,536	301,467	352,309	505,248	660,726	818,738	979,271
Cashflows at the End of Period	160,607	205,567	252,536	301,467	352,309	505,248	660,726	818,738	979,271	1,142,311

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 37 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	160,607	205,567	252,536	301,467	352,309	505,248	660,726	818,738	979,271	1,142,311
Inventory	10,843	11,169	11,504	11,849	12,204	12,570	12,948	13,336	13,736	14,148
Total assets traded	171,450	216,736	264,039	313,316	364,513	517,818	673,674	832,074	993,007	1,156,459
Non-current assets										
Fixed assets (net)	489,314	452,468	415,131	377,291	338,938	300,061	260,648	220,688	180,169	139,080
Total non-current assets	489,314	452,468	415,131	377,291	338,938	300,061	260,648	220,688	180,169	139,080
Total assets	660,764	669,204	679,170	690,607	703,451	817,879	934,322	1,052,762	1,173,176	1,295,539
Liabilities										
Current liabilities										
Payables	21,645	22,430	23,244	24,088	24,964	25,872	26,814	27,791	28,804	29,855
Remaining amount of Loan	69,780	76,396	83,638	91,567	(0)	(0)	(0)	(0)	(0)	0
Total current liabilities	91,426	98,826	106,882	115,655	24,964	25,872	26,814	27,791	28,804	29,855
Non-current liabilities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
long term loans	251,600	175,205	91,567	0	0	0	0	0	0	0
Total long - term liabilities	251,600	175,205	91,567	0	0	0	0	0	0	0
Total liabilities	343,026	274,030	198,449	115,655	24,964	25,872	26,814	27,791	28,804	29,855
Property rights										
Members Contributions	256,746	256,746	256,746	256,746	256,746	256,746	256,746	256,746	256,746	256,746
Compulsory reserve	6,099	13,843	22,398	31,821	42,174	53,526	65,076	76,823	88,763	100,894
Profit (loss)	54,893	124,586	201,579	286,386	379,568	481,735	585,686	691,403	798,864	908,044
Total equity	317,738	395,174	480,722	574,952	678,487	792,007	907,508	1,024,971	1,144,372	1,265,684
Total liabilities and equity	660,764	669,204	679,170	690,607	703,451	817,879	934,322	1,052,762	1,173,176	1,295,539

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 38 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	(641,864)	160,289	145,207	147,216	149,179	151,089	152,939	155,479	158,011	160,533	163,040	2,151,534
Discount Factor	1.00	0.80	0.71	0.64	0.57	0.51	0.46	0.41	0.36	0.33	0.29	0.29
Net Present Value for	(641,864)	128,054	103,686	93,957	85,099	77,036	69,699	63,332	57,528	52,240	47,422	625,792

Table below (40) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 39 payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(641,864)	160,289	145,207	147,216	149,179	151,089	152,939	155,479	158,011	160,533	163,040
Project Value Returned	641,864	481,575	336,367	189,152	39,973	)	(264,055)	(419,533)	(577,545)	738,078)	(901,118)
Payback Period (Year)	4	1	1	1	1	0	0	0	0	0	0

Table 40 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.88
Net Present Value for Cashflows	761,981
Payback Period	4
Internal Rate of Return	%27.99

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 761,981 Jordan Dinars considering that the project provides 17 Job Opportunities for the governorate residents.

Table 41 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%27.99	4	12.05%
Revenues declined by 10%	%22.46	5	12.05%
Operating Expenses Increased by 10%	%23.51	5	12.05%